Annex I. Public Debt Sustainability Analysis

Bosnia and Herzegovina's public debt remains low and sustainable based on an updated DSA even after the impact of the crisis. Alternative scenarios suggest that debt is sustainable under most adverse scenarios, except for a scenario based on a constant primary deficit at 2020 levels.

1. The assumptions are in line with the macroeconomic framework baseline. The budgetary position is expected to be in deficit in 2020 and to return to a surplus over 2021–25. A gradual increase in the average maturity of outstanding domestic debt is assumed. IFI financing reflects the favorable long-term financial terms currently provided by the EBRD, EIB, IMF and WB.

2. Debt dynamics improve over the medium term. Public debt is estimated to temporarily increase to 38 percent by end-2020, reflecting a sharp recession and large increase in public health-related spending. Over the medium-term, the public debt is expected to return to a declining path and reach 26 percent by end-2025. IFIs and bilateral officials will continue to provide external financing through project-related loans. Domestic debt is expected to be financed by banks, with longer average maturity if modernized debt strategies are adopted and implemented.

3. Baseline and alternative scenarios

- **The coverage of public debt is general government debt.** In the baseline scenario, the government is projected to assume some health sector debts in the context of the COVID crisis. In a downside scenario, possible additional contingent liabilities include the following: court judgments (1½ percent of GDP) and public enterprise debt (very tentatively estimated around 10 percent of GDP net exposure). In a severe and protracted shock, additional contingent risks could arise from the financial sector.

- **Baseline projections are consistent with the recent developments, but subject to significant uncertainty**. Despite the projected growth decline in 2020, a rapid recovery is expected in 2021. The budgetary primary surplus will follow a similar path. The projections include the payments for health arrears and bank debt, and settlements of intergovernmental arrears. These policies will lead to a gradual decline of the debt-to-GDP ratio until reaching about 26 percent of GDP by end-2025. Of course, a more protracted-than-expected slowdown of the EU and/or BiH economies would lead to a much lower GDP growth and primary surplus.

- **Under the historical scenario, the debt-to-GDP ratio would stay unchanged.** The paths of public debt and gross financing needs would deviate from the baseline, largely due to small primary deficits instead of surpluses under the baseline. The debt ratio would remain close to but below the end-2020 level of 38 percent for the projection period.

- **The constant primary surplus scenario seems unrealistic, assuming the COVID-19 shock is temporary.** With a constant primary deficit at 2020 levels (3.6 percent of GDP) throughout

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the projection period, the debt-to-GDP ratio is expected to increase steadily to about 50 percent in 2025. Such large primary deficits in the coming years seem unrealistic but also unsustainable. The gross financing needs would follow a similar increasing trend. Available resources would be limited due to the small investor basis, largely banks, and underdeveloped secondary bond market.

- **The real exchange rate shock has limited impact.** The debt-to-GDP ratio is projected to decline, even though at a slower pace than in the baseline scenario.

Heat Map

Debt level 1/	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability shock
Gross financing needs 2/	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability Shock
Debt profile 3/	Market Perception	External Financing Requirements	Change in the Share of Short-Term Debt	Public Debt Held by Non-Residents	Foreign Currency Debt

Source: IMF staff.
1/ The cell is highlighted in green if debt burden benchmark of 70% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.
2/ The cell is highlighted in green if gross financing needs benchmark of 15% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.
3/ The cell is highlighted in green if country value is less than the lower risk-assessment benchmark, red if country value exceeds the upper risk-assessment benchmark, yellow if country value is between the lower and upper risk-assessment benchmarks. If data are unavailable or indicator is not relevant, cell is white.
Lower and upper risk-assessment benchmarks are:
200 and 600 basis points for bond spreads; 5 and 15 percent of GDP for external financing requirement; 0.5 and 1 percent for change in the share of short-term debt; 15 and 45 percent for the public debt held by non-residents; and 20 and 60 percent for the share of foreign-currency denominated debt.

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Table 1. Bosnia and Herzegovina: Public Sector Debt Sustainability Analysis (DSA)

(In percent of GDP unless otherwise indicated)

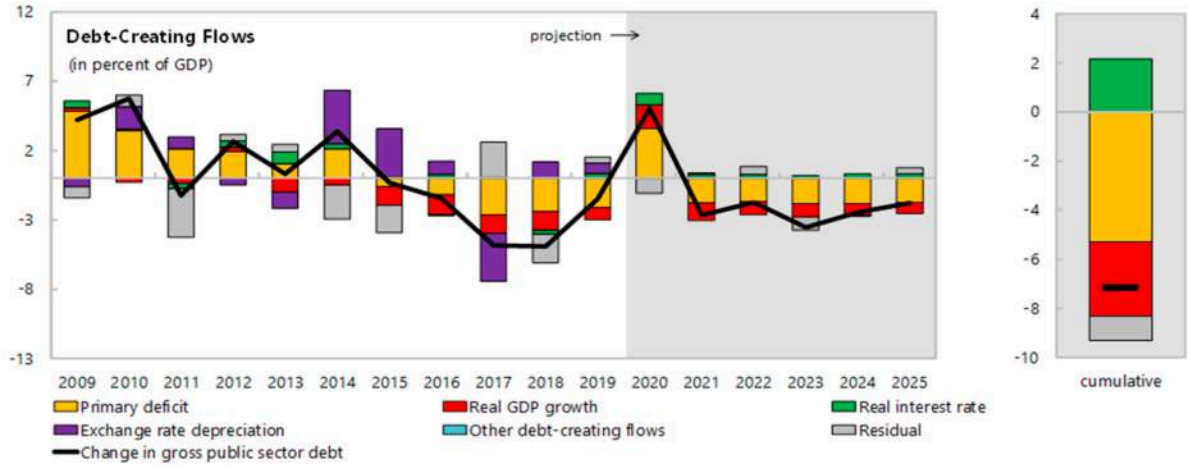
Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2011-2017 [2/]	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	42.7	34.3	32.8	37.9	35.2	33.5	29.9	27.5	25.7
Public gross financing needs	4.2	2.7	1.2	7.5	3.6	4.4	6.4	5.7	6.0
Real GDP growth (in percent)	1.9	3.6	2.7	-5.0	3.5	2.8	2.9	2.9	3.0
Inflation (GDP deflator, in percent)	1.2	2.8	1.1	-0.3	1.3	1.3	1.7	1.8	2.0
Nominal GDP growth (in percent)	3.1	6.5	3.8	-5.3	4.8	4.2	4.6	4.7	5.1
Effective interest rate (in percent) [4/]	1.8	2.0	2.2	2.1	2.1	2.2	2.3	2.9	3.3

As of December 31, 2019		
Sovereign Spreads		
EMBIG (bp) 3/		0
5Y CDS (bp)		0
Ratings	Foreign	Local
Moody's	B3	B3
S&Ps	B	B
Fitch	B	B

Contribution to Changes in Public Debt

	Actual			Projections							
	2011-2017	2018	2019	2020	2021	2022	2023	2024	2025	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	-0.2	-4.9	-1.5	5.0	-2.7	-1.8	-3.6	-2.4	-1.8	-7.1	-0.5
Identified debt-creating flows	0.4	-2.9	-1.9	6.1	-2.8	-2.4	-2.6	-2.3	-2.2	-6.2	
Primary deficit	0.4	-2.4	-2.1	3.6	-1.8	-1.7	-1.8	-1.8	-1.8	-5.3	
Primary (noninterest) revenue and grant	43.3	43.0	43.1	40.8	43.0	43.0	43.1	43.0	43.0	255.9	
Primary (noninterest) expenditure	43.7	40.6	41.0	44.3	41.3	41.3	41.3	41.2	41.2	250.6	
Automatic debt dynamics [5/]	0.0	-0.5	0.2	2.6	-1.0	-0.7	-0.7	-0.5	-0.5	-0.9	
Interest rate/growth differential [6/]	-0.5	-1.6	-0.5	2.6	-1.0	-0.7	-0.7	-0.5	-0.5	-0.9	
Of which: real interest rate	0.3	-0.3	0.4	0.8	0.3	0.3	0.2	0.3	0.3	2.2	
Of which: real GDP growth	-0.8	-1.3	-0.9	1.7	-1.3	-0.9	-0.9	-0.8	-0.8	-3.0	
Exchange rate depreciation [7/]	0.6	1.2	0.7	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
General Govt - Financing - Privatization	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	-0.6	-2.0	0.4	-1.1	0.1	0.6	-1.0	-0.1	0.4	-1.0	



Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as $[(r - \pi(1+g) - g + ae(1+r)]/(1+g+\pi+g\pi))$ times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as $r - \pi (1+g)$ and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as $ae(1+r)$.

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

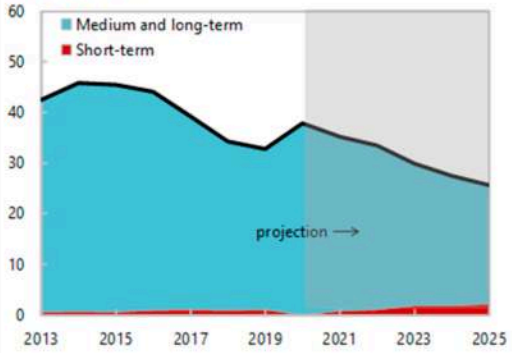
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Table 2. Bosnia and Herzegovina: Public DSA – Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

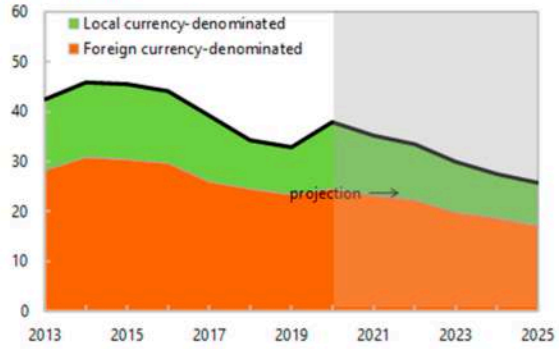
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)

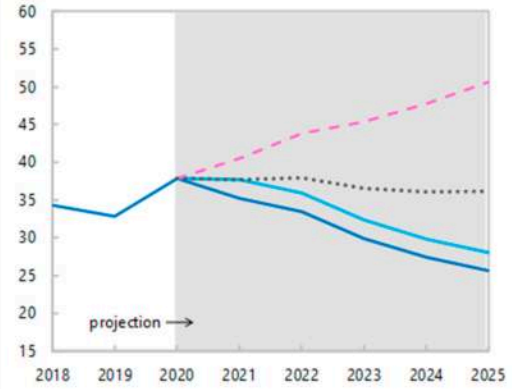


Alternative Scenarios



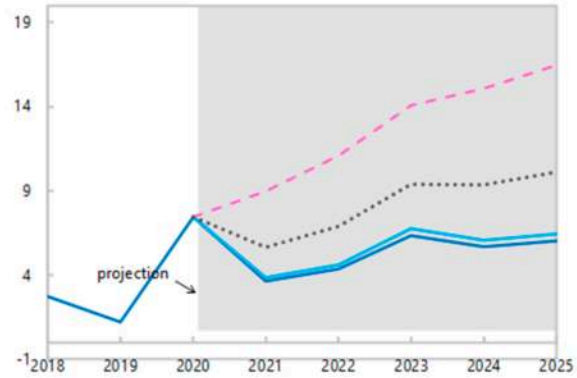
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-5.0	3.5	2.8	2.9	2.9	3.0
Inflation	-0.3	1.3	1.3	1.7	1.8	2.0
Primary Balance	-3.6	1.8	1.7	1.8	1.8	1.8
Effective interest rate	2.1	2.1	2.2	2.3	2.9	3.3
Constant Primary Balance Scenario						
Real GDP growth	-5.0	3.5	2.8	2.9	2.9	3.0
Inflation	-0.3	1.3	1.3	1.7	1.8	2.0
Primary Balance	-3.6	-3.6	-3.6	-3.6	-3.6	-3.6
Effective interest rate	2.1	2.1	2.1	2.1	2.4	2.5

Historical Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-5.0	2.0	2.0	2.0	2.0	2.0
Inflation	-0.3	1.3	1.3	1.7	1.8	2.0
Primary Balance	-3.6	-0.2	-0.2	-0.2	-0.2	-0.2
Effective interest rate	2.1	2.1	2.0	2.0	2.4	2.6
Real Exchange Rate Shock						
Real GDP growth	-5.0	3.5	2.8	2.9	2.9	3.0
Inflation	-0.3	5.8	1.3	1.7	1.8	2.0
Primary Balance	-3.6	1.8	1.7	1.8	1.8	1.8
Effective interest rate	2.1	2.3	2.2	2.3	2.9	3.3

Source: IMF staff.

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